

physics paper

Instructions: difficult paper

Section A

Attempt all questions

1. Derive the equation for the motion of a projectile under gravity, including all necessary assumptions and conditions.

Difficulty: Hard

Marks: 5

2. Explain the concept of wave-particle duality, including the experimental evidence that supports it.

Difficulty: Hard

Marks: 5

3. Discuss the limitations of the kinetic theory of gases, including the assumptions made and the range of validity.

Difficulty: Hard

Marks: 5

4. Derive the equation for the energy stored in a capacitor, including the necessary assumptions and conditions.

Difficulty: Hard

Marks: 5

5. Explain the concept of entropy, including its relationship to the second law of thermodynamics and its implications for the universe as a whole.

Difficulty: Hard

Marks: 5